Executive Summary: Titanic Survival Prediction Model

1. The Problem

The sinking of the RMS Titanic is one of the most infamous shipwrecks in history. The objective of this project was to build a predictive machine learning model to determine what sorts of people were more likely to survive the disaster. By framing this as a **Binary Classification** problem, we aimed to predict a passenger's survival status (Survived: 1, Did Not Survive: 0) based on their demographic and ticketing information, such as age, gender, passenger class, and fare.

2. Approach

Our methodology followed a rigorous end-to-end data science lifecycle:

- **Exploratory Data Analysis (EDA):** We analyzed the distributions and relationships of the variables. Key findings showed a stark disparity in survival rates: females survived at a much higher rate than males, and 1st-class passengers had significantly better outcomes than 3rd-class passengers.
- **Data Preprocessing:** * Handled missing values by imputing the median for numerical features (like Age) and the mode for categorical features (like Embarked).
    - Dropped highly collinear or structurally redundant features to prevent noise.
    - Applied Z-score standardization to numerical features (Age, Fare) and One-Hot Encoding to categorical features (Sex, Embarked).
- **Modelling:** We constructed `scikit-learn` Pipelines to prevent data leakage and evaluated three algorithms: **Logistic Regression** (Baseline), **Random Forest**, and a **Support Vector Classifier (SVC)**. We used 5-Fold Stratified Cross-Validation to ensure robust evaluation across different subsets of the data.

3. Results

The models were evaluated based on overall Accuracy, Precision, and Recall.

- **Model Comparison:** During cross-validation, both the Random Forest and the Support Vector Classifier significantly outperformed the Logistic Regression baseline.
- **Final Performance:** The **Support Vector Classifier (SVC)** was selected as the final model due to its stability. On an unseen 20% hold-out test set, the SVC achieved an **Accuracy of 81%**.
- **Insights:** The classification report indicated that the model is slightly better at predicting fatalities (higher recall for class 0) than survivals, which accurately reflects the imbalanced nature of the actual historical event where the majority of passengers did not survive.

4. Future Improvements

While the model performs well, several avenues could yield a higher accuracy and better predictive power:

1. **Advanced Feature Engineering:** Extracting titles (e.g., "Mr.", "Mrs.", "Master.") from the passenger names, or creating a "Family Size" feature by combining siblings/spouses and parents/children metrics, often provides strong predictive signals.
2. **Hyperparameter Tuning:** Implementing `GridSearchCV` or `RandomizedSearchCV` to fine-tune the internal parameters of the models (such as the `C` and `gamma` parameters in the SVC or tree depth in the Random Forest).
3. **Gradient Boosting:** Testing more sophisticated ensemble algorithms, such as **XGBoost** or **LightGBM**, which frequently achieve state-of-the-art results on tabular datasets like this one.
4. **Model Deployment:** Exporting the pipeline using `joblib` and integrating it into a lightweight web framework (like Flask or Streamlit) rather than a CLI to allow non-technical stakeholders to test predictions visually.